Supervised Machine Learning Approach for WSD in Spoken News

Sree Saicharan Vadapalli
ET23BTIT133

Evaluation Report

1 Setup

Everything here comes from one run of `src/pipeline.py`: 8 target words, 1,092 SemCor occurrences, a linear SVM and multinomial Naive Bayes, and window sizes $W = \pm 2$ and $W = \pm 5$. Scores are from stratified 5-fold cross-validation, so each occurrence is predicted once by a model that did not train on it. The Architecture and Methodology document covers the features and the protocol in detail.

2 Results

Table 1 has the per-word scores. "Majority" is the share of the most common sense, i.e. the accuracy you would get by always guessing it.

				SVM				Naive Bayes			
				Accuracy		Macro F1		Accuracy		Macro F1	
Word	n	Senses	Majority	±2	±5	±2	±5	±2	±5	±2	±5
bank	34	2	0.59	0.68	0.79	0.68	0.78	0.62	0.76	0.61	0.76
case	133	5	0.48	0.54	0.55	0.36	0.34	0.52	0.56	0.24	0.28
interest	140	7	0.41	0.61	0.59	0.47	0.37	0.54	0.50	0.23	0.27
plant	107	3	0.59	0.69	0.70	0.63	0.54	0.69	0.65	0.62	0.49
power	99	6	0.57	0.63	0.63	0.49	0.41	0.59	0.61	0.20	0.26
rate	116	3	0.59	0.88	0.85	0.82	0.74	0.84	0.83	0.70	0.57
run	185	12	0.36	0.49	0.48	0.31	0.29	0.46	0.46	0.16	0.19
state	278	6	0.39	0.72	0.71	0.57	0.52	0.68	0.67	0.45	0.44
Overall	1092			0.65	0.64	0.54	0.50	0.61	0.61	0.40	0.41

Table 1: Per-word results. The overall row is accuracy pooled over all occurrences and macro F1 averaged over words.

Both models beat the majority-sense guess on every word, and the SVM is the better of the two overall. The gap between accuracy and macro F1 is large for most words, which says the models get the common senses right and struggle with the rare ones.

3 Window size impact

A wider window does not help overall. SVM macro F1 drops from 0.54 to 0.50, mostly on *interest*, *plant*, *power* and *rate*, and Naive Bayes stays about flat. The useful cues for these words are local ("interest rate", "power plant"), and ±5 adds two to three times as many features, mostly rare words that act as noise at this data size.

bank is the exception (SVM accuracy 0.68 to 0.79). Its senses depend on topic, so words like "directors" and "approval" a few positions away help where the ±2 window only sees "of the" and "with the".

4 SVM vs Naive Bayes

The SVM is ahead on accuracy almost everywhere (only *case* at ±5 goes to Naive Bayes, 0.56 vs 0.55), but the real gap is macro F1: 0.54 vs 0.40 at ±2. On words with many senses Naive Bayes

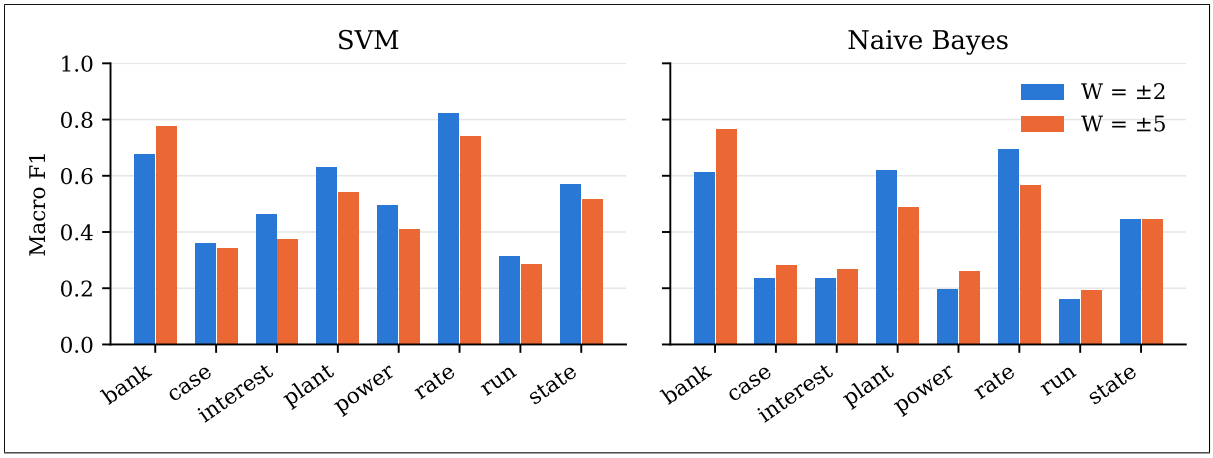


Figure 1: Macro F1 per word at both window sizes.

mostly predicts the majority sense (macro F1 of 0.16 on *run*), likely because its independence assumption double counts the overlapping POS and word features.

5 Failure analysis

Table 2 shows where the SVM goes wrong on *plant* at $W = \pm 2$.

	Predicted		
Gold	plant.n.01 (factory)	plant.n.02 (organism)	plant.v.01 (to plant)
plant.n.01 (factory)	50	13	0
plant.n.02 (organism)	15	21	0
plant.v.01 (to plant)	5	0	3

Table 2: SVM confusion matrix for *plant*, $W = \pm 2$.

Too little context. Many factory/organism errors are occurrences like "into each hole with the **plant**", where the ±2 window has no content words and the model falls back to the majority sense.

Rare senses get outvoted. 5 of the 8 verb uses of *plant* are predicted as a noun sense despite the verb POS tag. With about six verb examples per fold, the target POS does not carry enough weight. *interest.v.01* has the same problem (4 of 5 wrong).

Fine-grained senses. Some distinctions are hard even for people: "insurance **rates**" is gold *rate.n.02* (an amount charged) but predicted as *rate.n.01* (a magnitude). The financial sense of *interest* (*interest.n.04*), the one that matters most for news, is right only 2 times out of 14 and is usually mistaken for *interest.n.01* (curiosity).

6 Limitations

- SemCor is written text. Real transcripts have no punctuation or casing and contain recognition errors, which will hurt the POS tagger and through it the POS features. Expect lower numbers on actual spoken news.

- The data is small. *bank* has only 34 usable occurrences, so a difference of one prediction moves its accuracy by about 0.03.
- Both classifiers use default hyperparameters, and results are from a single cross-validation seed.

7 Conclusion

A linear SVM over target POS, neighbouring POS and a ±2 bag-of-words works best here, with 0.65 accuracy and 0.54 macro F1 overall. Widening the window to ±5 only pays off for words whose senses are topical, like *bank*; for the rest it adds noise. Naive Bayes is close on accuracy but much worse on rare senses. The main sources of error are short, uninformative windows, senses with very few examples, and WordNet distinctions that are finer than the context can resolve.